Topic 4 – Waves

Students should:		Maths skills
4.1	Recall that waves transfer energy and information without transferring matter	
4.2	Describe evidence that with water and sound waves it is the wave and not the water or air itself that travels	
4.3	Define and use the terms frequency and wavelength as applied to waves	
4.4	Use the terms amplitude, period, wave velocity and wavefront as applied to waves	
4.5	Describe the difference between longitudinal and transverse waves by referring to sound, electromagnetic, seismic and water waves	
4.6	Recall and use both the equations below for all waves: wave speed (metre/second, m/s) = frequency (hertz, Hz) × wavelength (metre, m) $v = f \times \lambda$ wave speed (metre/second, m/s) = distance (metre, m) ÷ time (second, s) $v = \frac{x}{t}$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
4.7	Describe how to measure the velocity of sound in air and ripples on water surfaces	2g
4.8P	**Calculate depth or distance from time and wave velocity**	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
4.9P	Describe the effects of a reflection b refraction c transmission d absorption of waves at material interfaces	5b
4.10	Explain how waves will be refracted at a boundary in terms of the change of direction **and speed**	1c 3c 5b
4.11	**Recall that different substances may absorb, transmit, refract or reflect waves in ways that vary with wavelength**	

Students should:		Maths skills
4.12P	**Describe the processes which convert wave disturbances between sound waves and vibrations in solids, and** **a explain why such processes only work over a limited frequency range** **b use this to explain the way the human ear works**	
4.13P	**Recall that sound with frequencies greater than 20 000 hertz, Hz, is known as ultrasound**	
4.14P	**Recall that sound with frequencies less than 20 hertz, Hz, is known as infrasound**	
4.15P	**Explain uses of ultrasound and infrasound, including** **a sonar** **b foetal scanning** **c exploration of the Earth's core**	1a, 1b, 1c, 2a 3a, 3b, 3c, 3d 5b
4.16P	Describe how changes, if any, in velocity, frequency and wavelength, in the transmission of sound waves from one medium to another are inter-related	1a, 1c, 1d 2a 3a, 3c, 3d
4.17	*Core Practical: Investigate the suitability of equipment to measure the speed, frequency and wavelength of a wave in a solid and a fluid*	2g

Use of mathematics

- Apply formulae relating velocity, frequency and wavelength (1c, 3c).
- Show how changes, if any, in velocity, frequency and wavelength, in transmission of sound waves from one medium to another, are inter-related (1c, 3c).

Suggested practicals

- Investigate models to show refraction, such as toy cars travelling into a region of sand.
- Investigate refraction in rectangular glass blocks.